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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which organic compound reacts with a Grignard reagent to give a product that forms a secondary alcohol on hydrolysis?

- ☐ **A** carbon dioxide
- ☐ **B** ethanal
- ☐ **C** methanal
- ☐ **D** propanone

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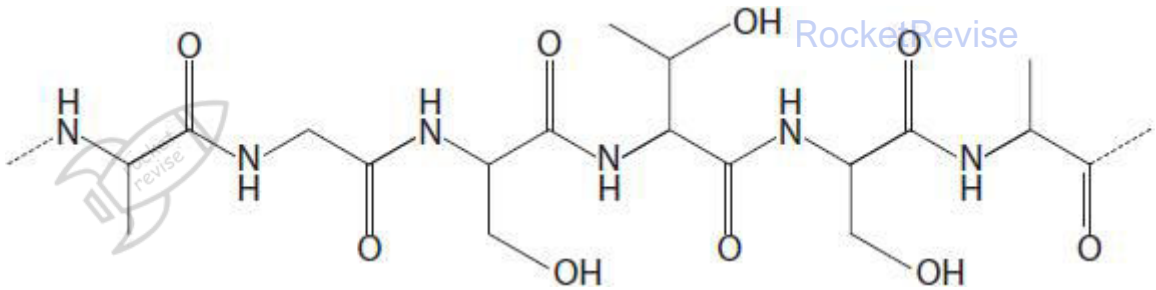
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Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Part of the structure of a protein is shown.



How many **different** amino acids were used to form this part of the structure?

- ☐ **A** 3
- ☐ **B** 4
- ☐ **C** 5
- ☐ **D** 6

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(Total for question = 1 mark)

Q3.

This question is about benzene and some related compounds.

(a) Some standard enthalpies of combustion are shown.

Compound	Structure	Standard enthalpy of combustion, $\Delta_c H^{\ominus}$ / kJ mol^{-1}
cyclohexene		–3752
cyclohexa-1,4-diene		–3584
benzene		–3267

(i) Using the standard enthalpies of combustion of cyclohexene and cyclohexa-1,4-diene, calculate a value for the enthalpy of combustion of the theoretical compound 'cyclohexa-1,3,5-triene'.

(2)



cyclohexa-1,3,5-triene

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(ii) Explain the difference between the enthalpy of combustion of 'cyclohexa-1,3,5-triene' calculated in (a)(i) and the enthalpy of combustion of benzene given in the table.

(3)

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(b) Bromine reacts with cyclohexene to form 1,2-dibromocyclohexane, and with benzene to form bromobenzene.

Compare and contrast these reactions, considering the type and mechanism of each reaction and the conditions required.
You are **not** required to draw the mechanisms of the reactions.

(4)

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(c) Bromine also reacts with phenol.

(i) Identify, by name or formula, the organic product when phenol reacts with **excess** bromine.

(1)

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(ii) Explain why bromine reacts much faster with phenol than with benzene.

(2)

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(Total for question = 12 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The reaction of ethene with bromine occurs under normal laboratory conditions but the reaction of benzene with bromine to form bromobenzene requires heat and the presence of a catalyst.

The best explanation for the difference in reactivity is that the delocalised electrons in benzene

☐ **A** repel electrophiles

☐ **B** result in a kinetic barrier to intermediate formation

☐ **C** result in benzene having an endothermic enthalpy of formation

☐ **D** make benzene thermodynamically stable with respect to the formation of bromobenzene

(Total for question = 1 mark)

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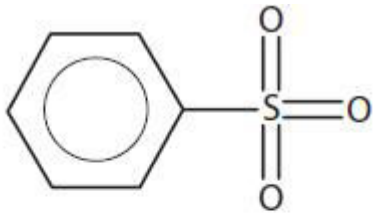
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Q5.

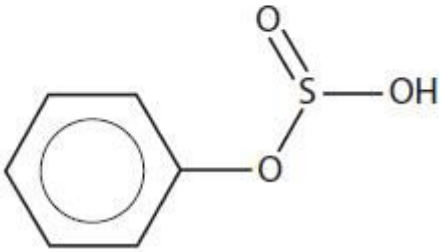
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

What is the product when benzene reacts with fuming sulfuric acid?

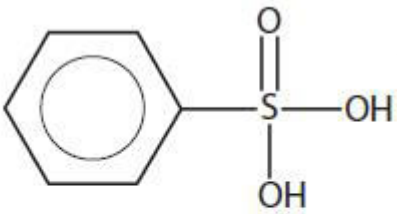
☐ **A**



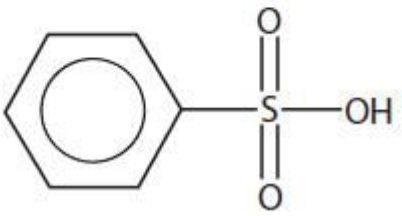
☐ **B**



☐ **C**



☐ **D**



(Total for question = 1 mark)

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Q6.

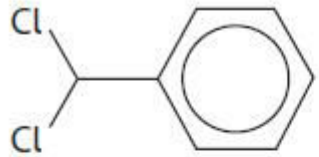
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these could **not** be formed when excess benzene is heated with trichloromethane, $CHCl_3$, in the presence of an aluminium chloride catalyst?

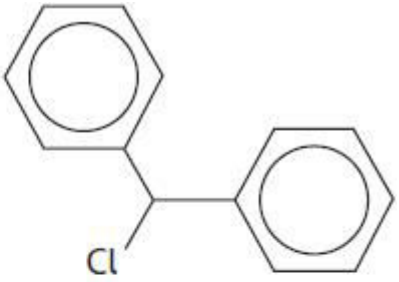
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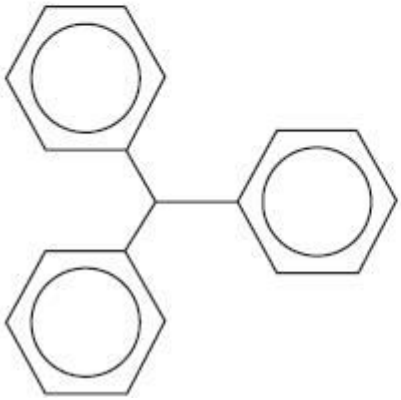
☒ **A**



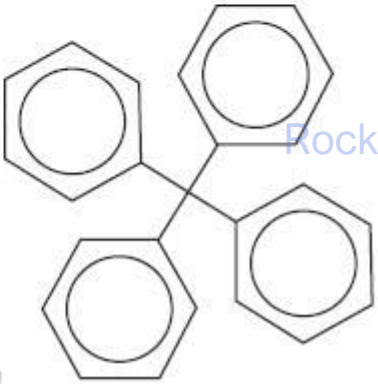
☒ **B**



☒ **C**



☒ **D**



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(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The delocalised electrons in benzene result from the overlap of

☒ **A** s orbitals to form σ bonds

☒ **B** s orbitals to form π bonds

☒ **C** p orbitals to form σ bonds

☒ **D** p orbitals to form π bonds

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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Benzene is sometimes represented by a Kekulé structure.



Kekulé structure of benzene

If this were the **only** structure of benzene, what would be the total number of isomers of dichlorobenzene?

- ☒ **A** two
- ☒ **B** three
- ☒ **C** four
- ☒ **D** five

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(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the molar mass, in g mol^{-1}, of the organic product when phenol reacts with **excess** bromine water?

- ☒ **A** 156.9
- ☒ **B** 172.9
- ☒ **C** 330.7
- ☒ **D** 488.5

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many σ bonds and π bonds are there in a molecule of benzene?

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	σ bonds	π bonds
☒ **A**	6	3
☒ **B**	6	6
☒ **C**	12	3
☒ **D**	12	6

(Total for question = 1 mark)

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Q11.

* Delocalised electron systems are important in determining the chemical properties of some compounds.

Compare and contrast the chemical reactions of bromine with benzene and cyclohexene, and with benzene and phenol, by considering the effects of delocalised electrons.

Detailed descriptions of the bonds or of the reaction mechanisms involved are **not** required.

(Total for question = 6 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Benzene reacts with fuming sulfuric acid to form benzenesulfonic acid.

Fuming sulfuric acid is

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- ☒ **A** sulfuric acid with a concentration of 98 %
- ☒ **B** pure sulfuric acid
- ☒ **C** concentrated sulfuric acid containing dissolved sulfur dioxide
- ☒ **D** concentrated sulfuric acid containing dissolved sulfur trioxide

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is correct for the reaction of bromine with phenol when compared with benzene?

		Reactivity of phenol	Electron density of the ring in phenol
☐	**A**	more reactive	higher electron density
☐	**B**	more reactive	lower electron density
☐	**C**	less reactive	higher electron density
☐	**D**	less reactive	lower electron density

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(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which statement explains why bromine reacts more readily with phenol than with benzene?

- ☐ **A** the O—H bond in phenol is polar
- ☐ **B** the oxygen atom in phenol is very electronegative
- ☐ **C** a lone pair of electrons on oxygen in phenol is delocalised into the ring
- ☐ **D** the electron density in the ring is greater in benzene than in phenol

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(Total for question = 1 mark)

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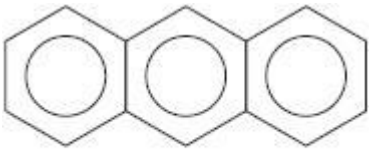
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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the **total** number of delocalised electrons in anthracene?



anthracene

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- ☐ **A** 6
- ☐ **B** 10
- ☐ **C** 14
- ☐ **D** 18

(Total for question = 1 mark)

Q16.

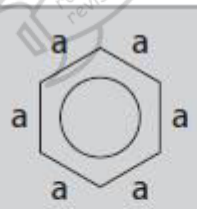
* Some thermochemical, X-ray diffraction and bromination information on benzene and cyclohexene is shown.

Thermochemical data

Compound	Enthalpy of hydrogenation / kJ mol^{-1}
benzene	−208
cyclohexene	−120

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X-ray diffraction data



Benzene bond	a
Bond length / pm	139



Cyclohexene bond	d	e	f
Bond length / pm	129	150	153

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Bromination

Compound	Reaction conditions	Organic product
benzene	$FeBr_3$, heat	bromobenzene
cyclohexene	room temperature	1,2-dibromocyclohexane

Explain how **all** this information provides evidence that the electrons in the π-bonds of benzene are delocalised.

(6)

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(Total for question = 6 marks)

Q17.

Benzene can be represented by either a cyclic triene or with a delocalised ring of electrons.





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*(a) Discuss the evidence, including one example from each of spectroscopy, thermochemistry and the type of reaction normally undergone, that supports the view that the better representation of benzene is with a delocalised ring of electrons.

(6)

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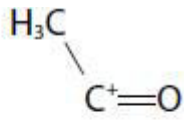
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(b) Benzene can be converted into phenylethanone by a Friedel-Crafts acylation.

(i) Complete the diagram, including curly arrows, to show the mechanism for this reaction.

(4)





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(ii) Write an equation to show how the species, CH_3CO^+, could be generated.

(1)

(c) Explain why phenol reacts with bromine more readily than benzene reacts with bromine.

(2)

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(d) Benzene can be converted into nitrobenzene.

(i) Complete the flow diagram showing this conversion.

(2)



(ii) Calculate the percentage yield if 0.642 g of nitrobenzene was made from 0.936 g of benzene.

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Give your answer to an appropriate number of significant figures.

(3)

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(Total for question = 18 marks)

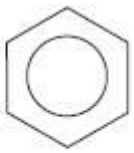
Q18.

In 1865, Friedrich August Kekulé suggested a structure for benzene which consisted of alternating single and double carbon–carbon bonds.



Kekulé structure

However, modern analytical techniques indicate a structure in which the electrons are delocalised.



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delocalised structure

(a) Explain how the results of X-ray diffraction experiments on benzene suggest a delocalised structure rather than the Kekulé structure.

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(b) The compound 1,2-dichlorobenzene exists as only one structure. Explain how this supports the delocalised structure of benzene rather than the Kekulé structure.

(2)

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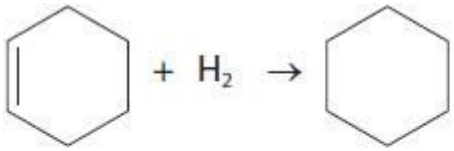
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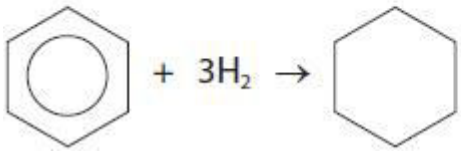
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(c) State how the enthalpy changes of hydrogenation for cyclohexene and benzene provide evidence for the delocalised structure.



$\Delta H = -118\,\text{kJ mol}^{-1}$



$\Delta H = -205\,\text{kJ mol}^{-1}$

(1)

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(d) Describe the structure of benzene in terms of the atomic orbitals involved, the bonds formed and the delocalised electrons.

(3)

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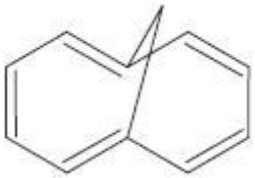
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(Total for question = 8 marks)

Q19.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An aromatic compound with a ten-carbon ring has the structure shown.



(a) Evidence for the aromaticity of this compound is that the carbon-carbon bonds in the ten-carbon ring

(1)

- ☐ **A** alternate in length between single and double bonds
- ☐ **B** all have similar lengths
- ☐ **C** have no regular pattern of bond length
- ☐ **D** are all similar to the carbon-carbon double bond length

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(b) The aromaticity is due to the formation of pi (π) bonds as a result of overlap of

(1)

- ☐ **A** s and p orbitals
- ☐ **B** p and d orbitals
- ☐ **C** p orbitals
- ☐ **D** d orbitals

(Total for question = 2 marks)

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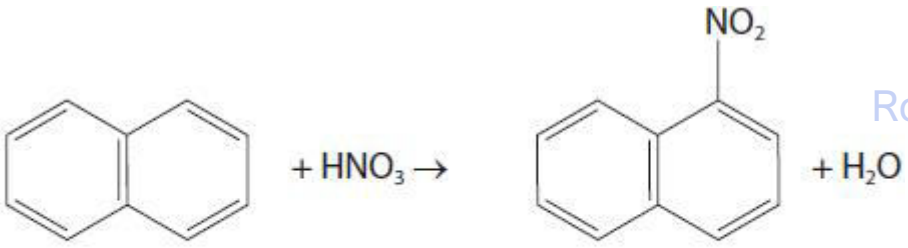
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Q20.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Naphthalene is an aromatic molecule with two aromatic rings joined together.

It can be nitrated at position 1 as shown.



naphthalene
Molar mass = 128 g mol^{-1}

1-nitronaphthalene
Molar mass = 173 g mol^{-1}

(a) In an experiment, a 2.00 g sample of naphthalene was nitrated and produced 2.57 g of 1-nitronaphthalene.

What is the percentage yield of 1-nitronaphthalene?

(1)

- ☒ **A** 95.1%
- ☒ **B** 77.8%
- ☒ **C** 74.0%
- ☒ **D** 57.6%

(b) A student carrying out this experiment calculated the percentage yield of the reaction to be greater than 100%.

What is the **most likely** explanation for this result?

(1)

- ☒ **A** there was unreacted naphthalene present
- ☒ **B** the naphthalene was impure
- ☒ **C** the sample of 1-nitronaphthalene was damp
- ☒ **D** other isomers of 1-nitronaphthalene were produced

(Total for question = 2 marks)

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Phenol reacts with bromine to form 2,4,6-tribromophenol.

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$C_6H_5OH + 3Br_2 \rightarrow C_6H_2Br_3OH + 3HBr$

[M_r values: C_6H_5OH = 94.0 Br_2 = 159.8 $C_6H_2Br_3OH$ = 330.7 HBr = 80.9]

(a) What is the percentage atom economy (by mass) for the formation of 2,4,6-tribromophenol?

(1)

- ☒ **A** 42.3%
- ☒ **B** 57.7%
- ☒ **C** 69.0%
- ☒ **D** 100%

(b) When 5.00 g of phenol was reacted and purified, the percentage yield of 2,4,6-tribromophenol was 76.8%.

What mass of purified 2,4,6-tribromophenol was formed?

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(1)

- ☒ **A** 3.84 g
- ☒ **B** 13.5 g
- ☒ **C** 17.6 g
- ☒ **D** 22.9 g

(Total for question = 2 marks)

Q21.

Q22.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The formation of nitrobenzene requires benzene and concentrated nitric and sulfuric acids.

Which is an equation for the reaction to form the electrophile?

- ☒ **A** $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + SO_4^{2-} + H_3O^+$
- ☒ **B** $HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + 2HSO_4^- + H_3O^+$
- ☒ **C** $HNO_2 + 2H_2SO_4 + H_2O \rightarrow NO_3^+ + 2HSO_4^- + 5H^+$
- ☒ **D** $HNO_3 + 3H_2SO_4 + OH^- \rightarrow NO_2 + 3HSO_4^- + H_3O^+ + H_2O$

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(Total for question = 1 mark)

Q23.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which type of data is **not** used as evidence for the structure or stability of the benzene ring?

- ☐ **A** infrared spectroscopy
- ☐ **B** mass spectrometry
- ☐ **C** thermochemical
- ☐ **D** X-ray diffraction

(Total for question = 1 mark)

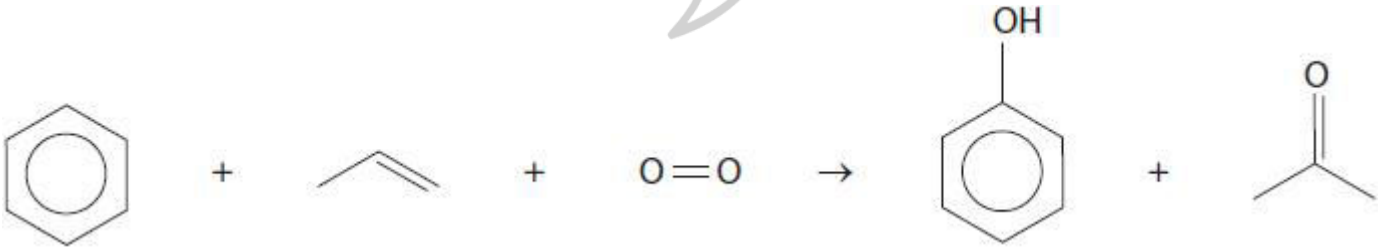
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Q24.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The cumene process is the main industrial method for making phenol.

The overall equation for this process is shown.



What is the atom economy by mass of the cumene process for making phenol?

- ☐ **A** 61%
- ☐ **B** 62%
- ☐ **C** 94%
- ☐ **D** 95%

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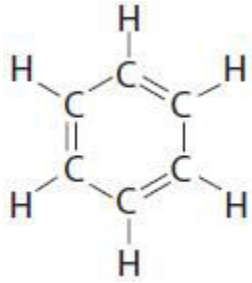
(Total for question = 1 mark)

Q25.

In 1865, Kekulé proposed a cyclic structure for benzene, C_6H_6.

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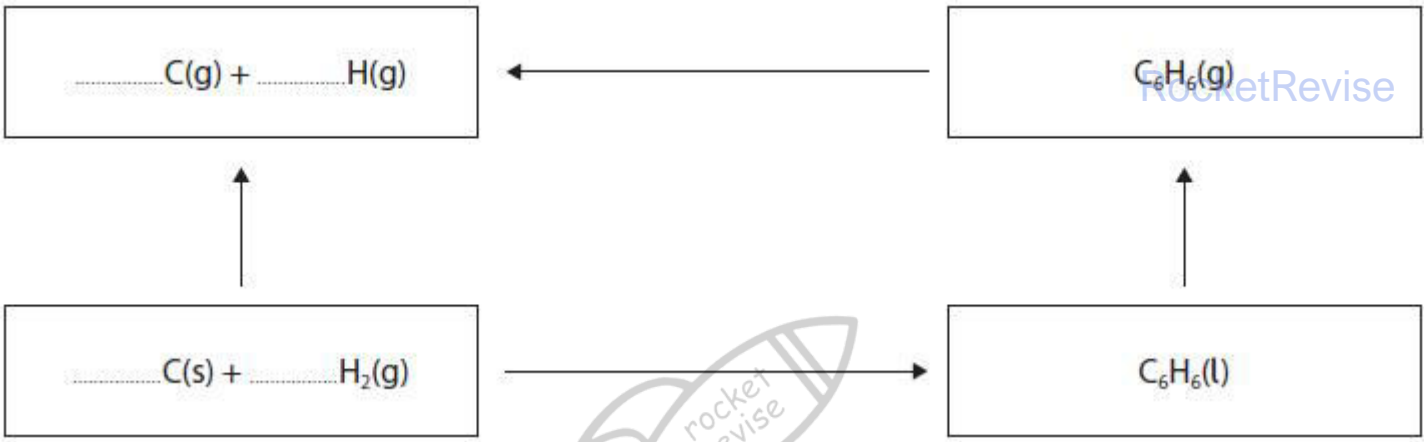
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(a) An incomplete Hess cycle is shown.

(i) Complete the cycle by adding balancing numbers.

(1)



(ii) Calculate the mean carbon-carbon bond enthalpy in benzene, using your completed cycle and the data shown.

(2)

$C_6H_6(l)$ $\Delta_f H = +49.0\,kJ\,mol^{-1}$

$C(s) \rightarrow C(g)$ $\Delta_{at} H = +715\,kJ\,mol^{-1}$

$H_2(g) \rightarrow 2H(g)$ $\Delta_{at} H = +436\,kJ\,mol^{-1}$

$C_6H_6(l) \rightarrow C_6H_6(g)$ $\Delta_r H = +31\,kJ\,mol^{-1}$

Bond enthalpy C—H = $+413\,kJ\,mol^{-1}$

(iii) Explain why the bond enthalpy value calculated in (a)(ii) is not the exact mean of the carbon-carbon double bond and carbon-carbon single bond enthalpies that would be expected from the Kekulé structure.

Bond enthalpy C—C = +347 kJ mol^{-1}
Bond enthalpy C=C = +612 kJ mol^{-1}

(2)

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(b) Describe how all the carbon-carbon bonds in benzene are formed.

(2)

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(c) State one chemical property and one physical property of benzene which would **not** be shown by the compound with the Kekulé structure.

(2)

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(Total for question = 9 marks)

Q26.

* Compare and contrast the bromination of benzene and the bromination of phenol.

Include equations in your answer.

Detailed mechanisms are **not** required.

(Total for question = 6 marks)

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